

iAM Smart API Specification FOR CORPID INTEGRATION

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Document Revision History

Ver. No.	Date	Section Affected	Summary of Changes
0.1.0	30 March 2025	All	Initial draft
0.1.1	20 May 2025	1.4	1.4.2 Update callback response; add 1.4.3 reference; add 1.4.4-1.4.7 for more callback types

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1. AUTHENTICATION

1.1 Overview

E-service can make use of the “iAM Smart” API and CorpID to verify the identities of the users and login users in a simple and secure way.

1.2 Scopes

E-service shall apply the following scope values to “iAM Smart” Support team in advance and provide the required scope values based on the business requirement specified in the API request. In addition, e-service can check the approved scope values in the Self-Service Portal (ESP).

Scope	Description
eidapi_auth	Scope for Authentication

1.3 Use Cases and Scenarios

1.3.1 Authentication (E-service Website in Different Device)

The sequence diagram below shows how an e-service website leverages the “iAM Smart” System and the mock CorpID system to perform user authentication when the “iAM Smart” Mobile App is installed in a separated mobile device.

CorpID Sandbox Authentication (e-Service Website in Different Device)

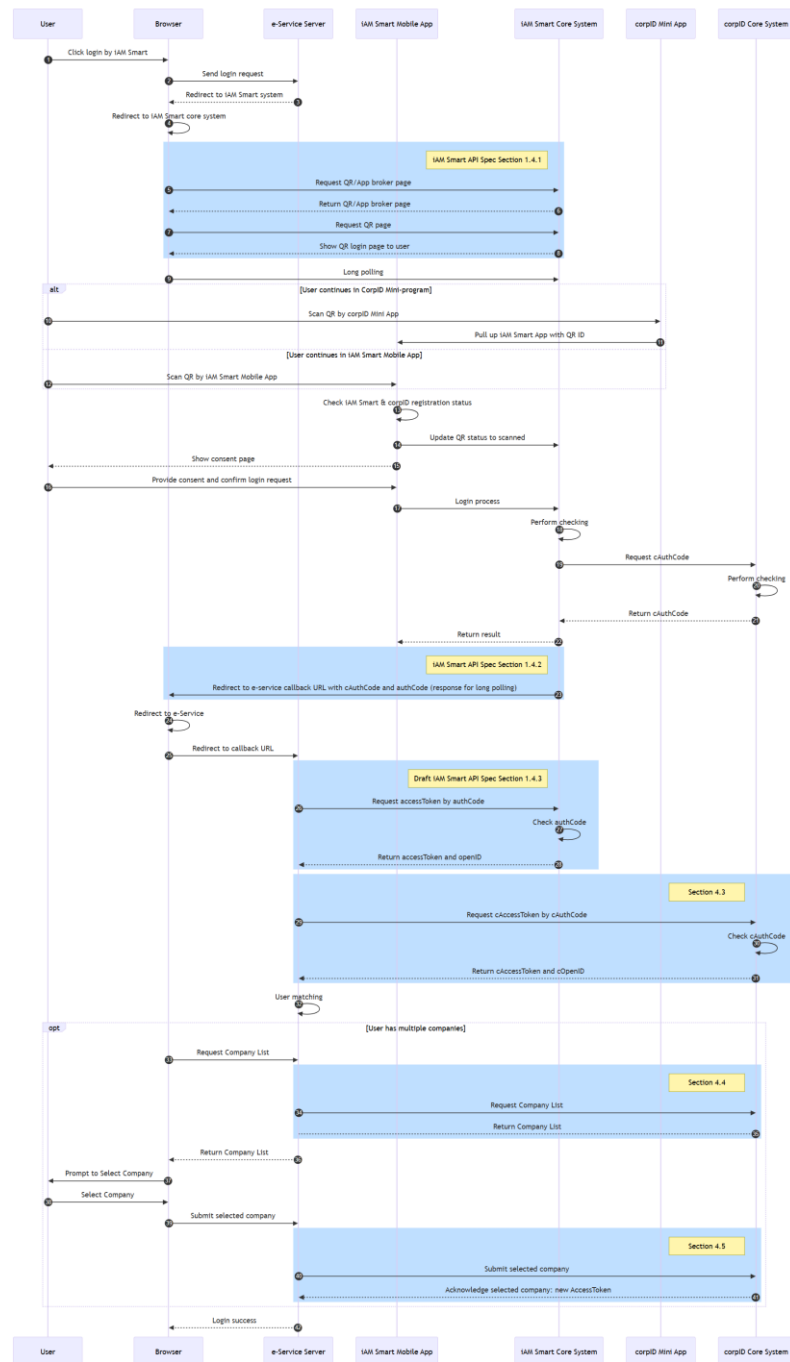


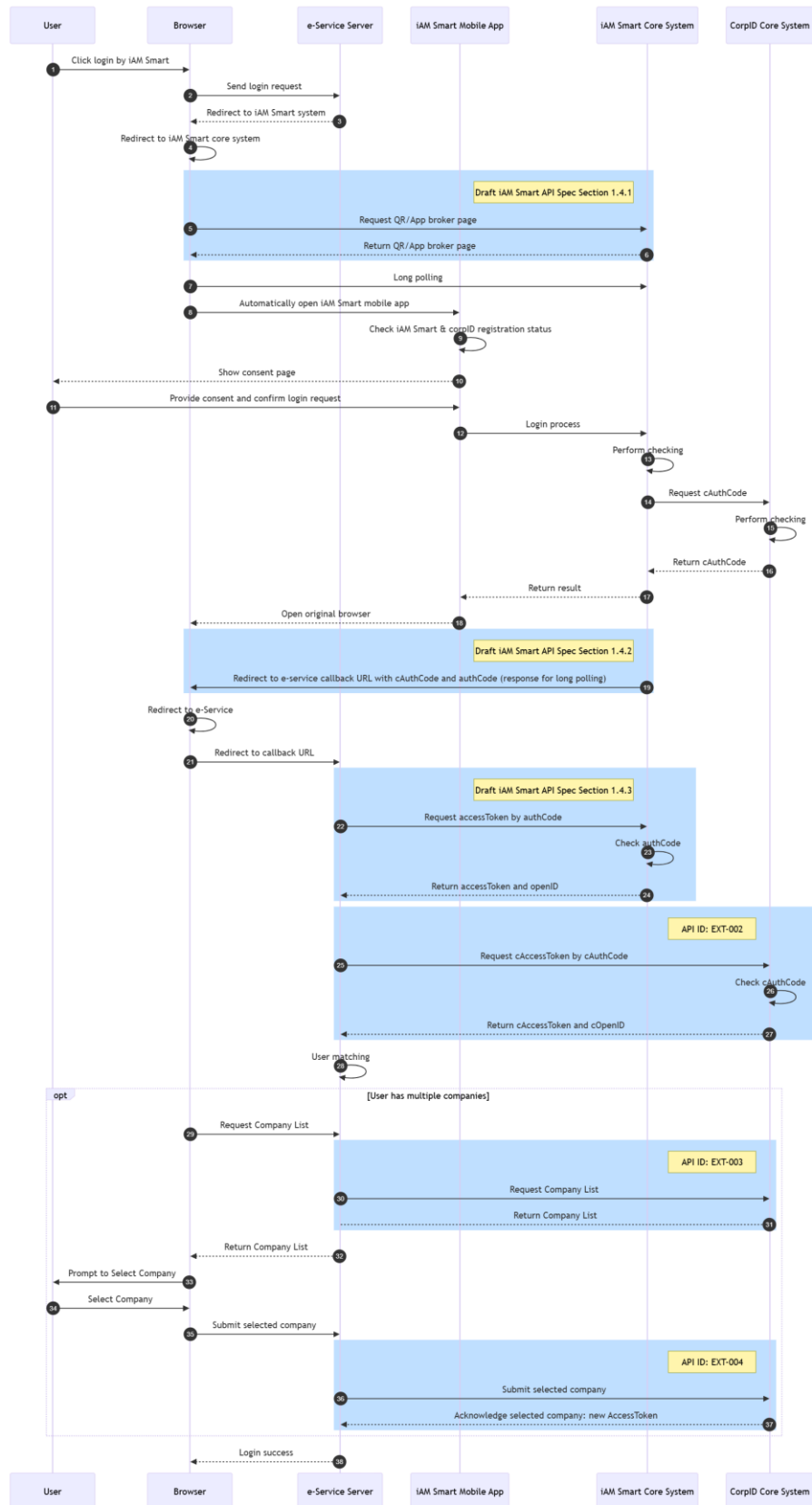
Figure-1 Authentication (E-service Website in Different Device)

APIs interactions between e-service, the mock CorpID system, and the “iAM Smart” System are marked in red. Technical details of respective APIs can be found in the following sections.

No.	API Name	API Reference (Section)
1	Request QR/App Broker Page	1.4.1
2	Redirect to e-service callback URL	1.4.2
3	Get iAM Smart Access Token by authCode	1.4.3
4	Get CorpID cAccessToken by cAuthCode	CorpID API Spec Section 4.2
5	[Optional] Request Corporation List	CorpID API Spec Section 4.3
6	[Optional] Request Token Exchange for Switching Corporation	CorpID API Spec Section 4.4

1.3.2 Authentication (E-service Website in Same Device)

CorpID Authentication (e-Service Website in Same Device)



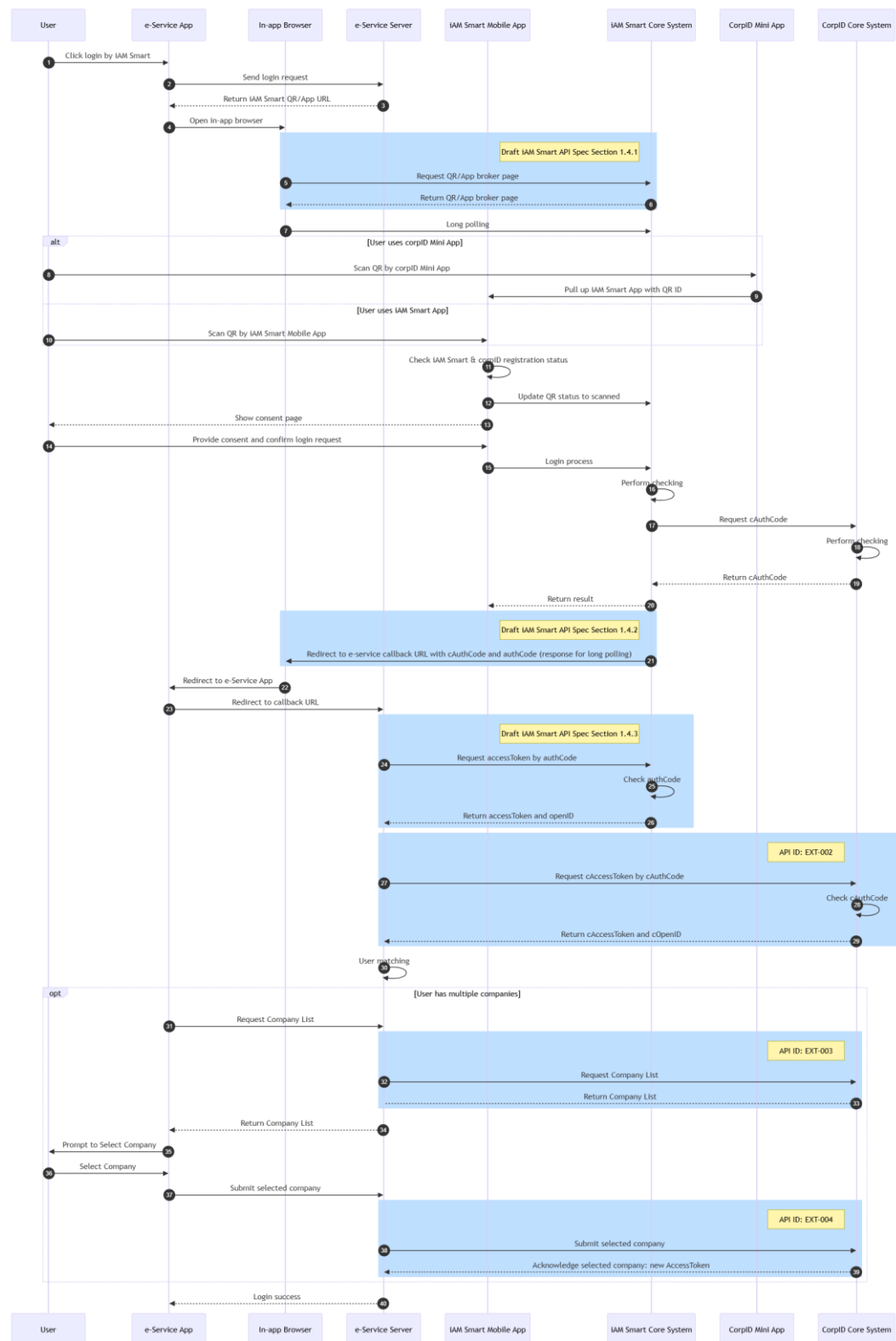
API interactions between e-service servers and CorpID System are listed below.

Technical details of respective APIs can be found in the following sections.

No.	API Name	API Reference (Section)
1	Request QR/App Broker Page	1.4.1
2	Redirect to e-service callback URL	1.4.2
3	Get iAM Smart Access Token by authCode	1.4.3
4	Get CorpID cAccessToken by cAuthCode	CorpID API Spec Section 4.2
5	[Optional] Request Corporation List	CorpID API Spec Section 4.3
6	[Optional] Request Token Exchange for Switching Corporation	CorpID API Spec Section 4.4

1.3.3 Authentication (E-service App in Different Device)

CorpID Authentication (e-Service App in Different Device)

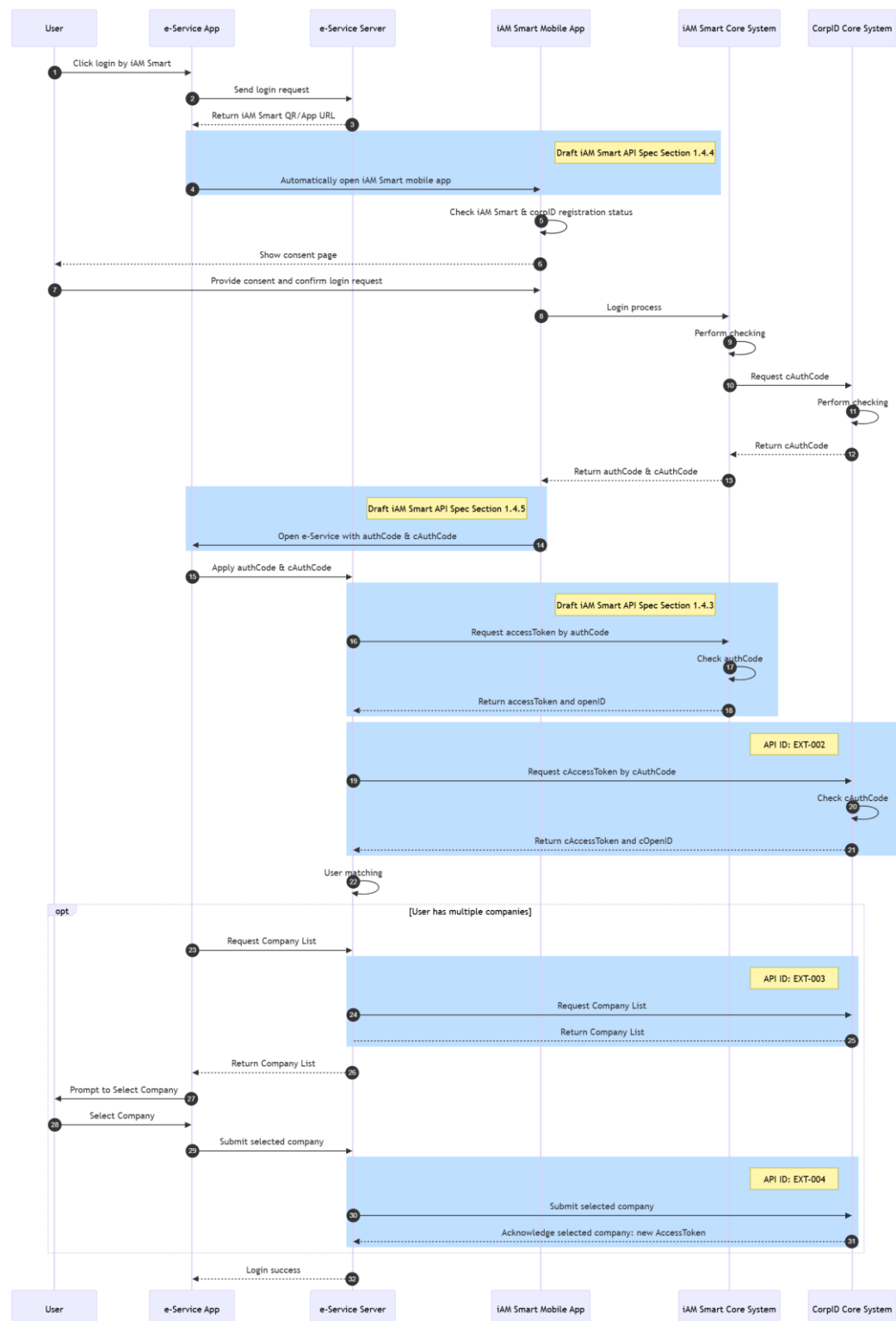


API interactions between e-service servers and CorpID System are listed below. Technical details of respective APIs can be found in the following sections.

No.	API Name	API Reference (Section)
1	Request QR/App Broker Page	1.4.1
2	Redirect to e-service callback URL	1.4.2
3	Get iAM Smart Access Token by authCode	1.4.3
4	Get CorpID cAccessToken by cAuthCode	CorpID API Spec Section 4.2
5	[Optional] Request Corporation List	CorpID API Spec Section 4.3
6	[Optional] Request Token Exchange for Switching Corporation	CorpID API Spec Section 4.4

1.3.4 Authentication (E-service App in Same Device)

CorpID Authentication (e-Service App in Same Device)



API interactions between e-service servers and CorpID System are listed below. Technical details of respective APIs can be found in the following sections.

No.	API Name	API Reference (Section)
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1	Open iAM Smart Mobile App for Authentication (appv2)	1.4.4
2	Call with cAuthCode to Online Service App	1.4.5
3	Get iAM Smart Access Token by authCode	1.4.3
4	Get CorpID cAccessToken by cAuthCode	CorpID API Spec Section 4.2
5	[Optional] Request Corporation List	CorpID API Spec Section 4.3
6	[Optional] Request Token Exchange for Switching Corporation	CorpID API Spec Section 4.4

1.4 API Implementation Details (iAM Smart)

1.4.1 Request QR Page

● API Description

Name	Description
Service Full Name	Request QR Page
URI (as in RESTFUL API)	<code>https://<iAM_Smart_domain>/api/v1/auth/corpId/getQR</code>
Request Type	GET
Service Version	1.0.0
Description of Service	E-service calls this API to get the QR/App broker page or QR page. After the user authorises login on the “iAM Smart” Mobile App, the page will be redirected to the <code>redirectURI</code> with <code>authCode</code> , <code>cAuthCode</code> and <code>state</code> parameters. If the user denies it, only the <code>state</code> parameter will be redirected.

● Request Parameters

Parameter	Type	Presence	Description
<code>clientID</code>	String	Required	E-service client identifier. The <code>clientID</code> will be assigned to e-service at the initial registration. Please refer to the self-service portal.
<code>responseType</code>	String	Required	The value MUST be set to <code>code</code> .
<code>source</code>	String	Required	Request initiator. The supported source values can be found in Appendix B of the “iAM Smart” API specification. Except (<code>App_Link</code> and <code>App_Scheme</code>).

redirectURI	String	Required	The callback redirect URI. The value should be URL encoded and registered in the self-service portal.
scope	String	Required	The value should be URL encoded. Specify the scope of “iAM Smart” functions: - eidapi_auth - eidapi_profiles - eidapi_formFilling - eidapi_sign - eidapi_fr - eidapi_bulksign The value of the scope parameter is expressed as a list of space-delimited, case-sensitive strings.
cScope	String	Required	The value should be URL encoded. Specify the scope of “CorpID” functions please reference to CorpID API Spec Section 2.5. The value of the scope parameter is expressed as a list of space-delimited, case-sensitive strings.
lang	String	Optional	Language to display: en-US, zh-HK, or zh-CN. If this parameter is not specified, zh-HK will be shown.
state	String	Optional	If the state parameter is presented in the request message, the same state value will be returned to e-service during the callback. It is used to prevent the CSRF attack. The value of state is defined by e-service and it should be a secure random string of any length less than or equal to 36 (UUID string length). Only ASCII letters, numbers, underscore and hyphens are accepted.
brokerPage	Boolean	Optional	If brokerPage is set to true, Universal Link (iOS) / App Link (Android) will be leveraged to open the “iAM Smart” Mobile App. This feature is useful for e-service supporting

			mobile web versions while triggering the “iAM Smart” Mobile App or showing a QR page automatically. (i.e. Show QR page without detecting whether the “iAM Smart” Mobile App is installed). The default value is <code>false</code> .
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● Example Request

```
// Line breaks are for legibility only.
GET
https://<iAM_Smart_domain>/api/v1/auth/corpid/getQR
?clientId=E-service1
&responseType=code
&source=Chrome
&redirectURI=https%3A%2F%2Frp_domain%2Frp_context%2Fcall_back_endpoint
&scope=eidapi_auth%20eidapi_formFilling%20eidapi_profiles
&lang=en-US
&state=eb9b7b8eddd5
```

● Response Parameters

N/A

1.4.2 Callback with cAuthCode to Online Service Server

● API Description

Name	Description
Service Full Name	Callback with cAuthCode to Online Service Server
URI (as in RESTFUL API)	https://<rp_domain>/<rp_context>/<call_back_endpoint>
Request Type	GET
Service Version	V1.0.0
Description of Service	This callback is used to pass <code>cAuthCode</code> and/or <code>authCode</code> to online service Server. The URI must be registered in the self-service portal.
API User	iAM Smart Broker Page
API Provider	Corpid Server

● **Request Parameters**

Parameter	Timeout value		Description
Callback	18 minutes	Required	Online service client identifier. The clientID will be assigned to online service at the initial registration. Please refer to the self-service portal.

● **Request Parameters**

Parameter	Type	Presence	Description
businessID	String	Required (Conditional)	businessID will only be returned in anonymous workflows. businessID is a unique identifier for online service to differentiate different requests. Online service can use the businessID to relate the callback message with the original request. Maximum Length: 36
code	String	Required (Conditional)	The authorisation code generated by the “iAM Smart” System. The authorisation code will be expired in 30 seconds after issuance. Online service MUST NOT use the authorisation code more than once. An error message will be returned if an authorisation code is expired or re-used.
cCode	String	Required (Conditional)	The authorisation code is generated by the “Corp ID” System. The authorisation code will be expired in 30 seconds after issuance. Online Service MUST NOT use the authorisation code more than once.

			An error message will be returned if an authorisation code is expired or re-used.
state	String	Optional	If the state parameter has been presented in the request message, the same state value will be returned. It is used to prevent the CSRF attack. The value of state is defined by online service and it should be a secure random value.
error_code	String	Required (Conditional)	Return error code when exception.

● Example Callback

```
// Line breaks are for legibility only.
GET
https://<call_back_endpoint>
?code=0ad186353c424c64897fcc00445c9ba1
&cCode=12340ad186353c424c64897fcc00445c
&state=eddd527b6
```

1.4.3 Request iAM Smart accessToken & Tokenised ID

* Please refer to ST13-iAM Smart API Specification v2.5.0, API 3.4.5 Request accessToken & Tokenised ID.

1.4.4 Open iAM Smart Mobile App for Authentication (appv2)

● Using URL Scheme

Name	Description
Service Full Name	Open iAM Smart Mobile App for authentication
URI (as in RESTFUL API)	<"iAM Smart" app URL scheme>://v2_cauth
Service Version	V2.0.0
Description of Service	The URL scheme makes use of deep linking to redirect users to process authentication action in the "iAM Smart" Mobile App and/or "Corp ID" Mini App. The URL schemes are supported on the iOS and Android versions of the "iAM Smart" Mobile App.
API User	Online Service App
API Provider	iAM Smart App

● Request Parameters

Parameter	Type	Presence	Description
clientID	String	Required	Online service client identifier. The clientID will be assigned to online service at the initial registration.
responseType	String	Required	the value MUST be set to code.
source	String	Required	App_Link (iOS) or App_Package (Android)
clientRedirectURI	String	Required (iOS only)	Callback redirect URI. The value should be URL encoded and registered in the self-service portal.
packageName	String	Required (Android only)	Package name of the Online Service App for callback use.
activityClass	String	Required (Android only)	Activity class name of the Online Service App for callback use.
activityParams	String	Optional (Android only)	Optional params used during callback.
scope	String	Required	The value should be URL encoded. Specify the scope of "iAM Smart" functions: - eidapi_auth - eidapi_profiles - eidapi_formFilling

			- eidapi_sign - eidapi_fr - eidapi_bulksign The value of the scope parameter is expressed as a list of space-delimited, case-sensitive strings.
cScope	String	Required	The value should be URL encoded. Specify the scope of “corp ID” functions: - corpidapi_auth - corpidapi_profiles - corpidapi_formFilling - corpidapi_sign - corpidapi_fr - corpidapi_bulksign (For reference only, the final scope will be provided by the corpID team) The value of the scope parameter is expressed as a list of space-delimited, case-sensitive strings.
state	String	Optional	If state parameter is presented in the request message, the same state value will be returned to online service during callback. It is used to prevent the CSRF attack. The value of state is defined by online service and it should be a secure random string of any length less than or equal to 36 (UUID string length). Only ASCII letters, numbers, underscore and hyphens are accepted.

● Example Scheme

```
// Line breaks are for legibility only.
<"iAM Smart" app URL scheme>://v2_cauth
?clientID=Online Service1
&responseType=code
&source=App_Package
&packageName=com.onlineservice.myapp
&activityClass=callback_activity
&activityParams=callback_param
&scope=eidapi_auth%20eidapi_formFilling%20eidapi_profiles
```

```
&cScope=corpapi_auth%20corpapi_formFilling%20corpapi_profiles
&state=eb9b7b8eddd5
// Note: The values in scopes and/or cScopes are space-delimited with URL Encoded.
```

1.4.5 Callback with cAuthCode to Online Service App

● API Description

Name	Description
Service Full Name	Callback with cAuthCode to Online Service App
Universal Link (iOS only)	The online service custom app Universal Link should be registered with the “iAM Smart” System during onboarding.
Package name (Android only)	<package name> and <activity class name>. The online service package name must be registered in the self-service portal.
Description of Service	For appv2 API, the Android Online Service App will be invoked and launched by the package name. It make use of intent to redirect users to the Online Service app. The iOS Online Service App will be invoked and launched via Universal link. The Universal Link with the landing location as well as the package name plus activity class name must be registered in the self-service portal and enabled by the support team.
API User	iAM Smart App
API Provider	Online Service App

● Request Parameters

Parameter	Timeout value		Description
Callback	18 minutes	Required	Online service could treat the request as failed if it doesn't get callback within 18 minutes.

● Request Parameters

Parameter	Type	Presence	Description
businessID	String	Required (Conditional)	businessID will only be returned in anonymous workflows. businessID is a unique identifier for online service to differentiate different

			requests. Online service can use the <code>businessID</code> to relate the callback message with the original request. Maximum Length: 36
<code>code</code>	String	Required (Conditional)	The authorisation code generated by the “iAM Smart” System. The authorisation code will be expired in 30 seconds after issuance. Online service MUST NOT use the authorisation code more than once. An error message will be returned if an authorisation code is expired or re-used.
<code>cCode</code>	String	Required (Conditional)	The authorisation code is generated by the “Corp ID” System. The authorisation code will be expired in 30 seconds after issuance. Online Service MUST NOT use the authorisation code more than once. An error message will be returned if an authorisation code is expired or re-used.
<code>state</code>	String	Optional	If the <code>state</code> parameter has been presented in the request message, the same state value will be returned. It is used to prevent the CSRF attack. The value of state is defined by online service and it should be a secure random value.
<code>error_code</code>	String	Required (Conditional)	Return error code when exception.

● Example Universal Link

```
// Line breaks are for legibility only.
<Universal Link>://<landing location>
?code=0ad186353c424c64897fcc00445c9ba1
?cCode=221bd186353c424cccc97fcc00445c9b
&state=eddd527b6
```

● Example Package Name

```
Intent ii = new Intent(<package name>, <activity class name>);
ii.putExtra("code", "0ad186353c424c64897fcc00445c9ba1");
ii.putExtra("cCode", "221bd186353c424cccc97fcc00445c9b");
ii.putExtra("state", "eddd527b6");
startActivity(ii);
```

1.4.6 Callback with cAuthCode to Online Service Server (Direct Login v2 to Web)

● API Description

Name	Description
Service Full Name	Callback with cAuthCode to Online Service Server for direct login v2 of Service Catalogue
URI (as in RESTFUL API)	https://<rp_domain>/<rp_context>/<call_back_endpoint>
Request Type	GET
Service Version	V2.0.0
Description of Service	This callback is used to pass cAuthCode to online service Server. The online service MUST hold a separate callback endpoint to receive the cAuthCode for Direct Login V2. The URI must be registered in the self-service portal (ESP) and approved by the support team, and online service can view the configuration in the ESP.
API User	iAM Smart Broker Page
API Provider	CorpID Server

● Callback Parameters

Parameter	Timeout value		Description
code	String	Required	The authorisation code is generated by the “iAM Smart”

		(Conditional)	System. The authorisation code will be expired in 30 seconds after issuance. Online Service MUST NOT use the authorisation code more than once. An error message will be returned if an authorisation code is expired or re-used.
cCode	String	Required (Conditional)	The authorisation code is generated by the “Corp ID” System. The authorisation code will be expired in 30 seconds after issuance. Online Service MUST NOT use the authorisation code more than once. An error message will be returned if an authorisation code is expired or re-used.
RefID	String	Required (Conditional)	A unique value for troubleshooting when necessary.
error_code	String	Required (Conditional)	Return error code to online service pre-defined fallback page when exception occurred.

- **Example Callback**

```
// Line breaks are for legibility only.
GET
https://<call_back_endpoint>
?code=0ad186353c424c64897fcc00445c9ba1
&cCode=12340ad186353c424c64897fcc00445c
&RefID=eddd527b6
```

1.4.7 Callback with cAuthCode and code_verifier to Online Service App (Direct Login v2 to App)

- **URL Scheme and package name**

Name	Description
------	-------------

Service Full Name	Callback with cAuthCode and code_verifier to Online Service App
Universal Link (iOS only)	The online service Universal Link should be registered with the “iAM Smart” System during onboarding.
Package name (Android)	<package name> and <activity class name>. The online service package name and class name should be registered with the “iAM Smart” System during onboarding.
Description of Service	This callback is used to pass cAuthCode and code_verifier to online service App.
API User	iAM SmartApp
API Provider	Online Service App

● **URL Scheme / activity Parameters**

Parameter	Timeout value		Description
code	String	Required (Conditional)	The authorisation code is generated by the “iAM Smart” System. The authorisation code will be expired in 30 seconds after issuance. Online Service MUST NOT use the authorisation code more than once. An error message will be returned if an authorisation code is expired or re-used.
cCode	String	Required (Conditional)	The authorisation code is generated by the “Corp ID” System. The authorisation code will be expired in 30 seconds after issuance. Online Service MUST NOT use the authorisation code more than once. An error message will be returned if an authorisation code is expired or re-used.
code_verifier	String	Required	The PKCE code verifier generated by the “iAM Smart” System.

● **Example Universal Link**

```
<Universal Link>://<landing location>?code=0ad186353c424c64897fcc00445c9ba1
```

```
&cCode=12340ad186353c424c64897fcc00445c  
&code_verifier=eddd527b6
```

● Example Package Name

```
Intent ii=new Intent(<package name>, <activity class name>);  
ii.putExtra("code", "0ad186353c424c64897fcc00445c9ba1");  
ii.putExtra("cCode", "12340ad186353c424c64897fcc00445c");  
ii.putExtra("code_verifier", "DxjdFp0vKYCY4F0DE6M3eEadLxcJBTb6k4LZ9J5z");  
startActivity(ii);
```